The reason we do this: Variance Risk Premium

The entire point of this section is to convince you that the variance risk premium exists, will continue to exist, and that the way we use it to build the engine behind our strategies is going to put dollars in your pocket.

What is the Variance Risk Premium (VRP)?

The variance risk premium is the tendency for implied volatility to be higher than the subsequently realized volatility. It can be found across most assets: equity indices, the VIX, bonds, commodities, currencies and many individual stocks. It’s measured as the difference between implied volatility and the actual volatility that occurs over time.

In plain english, ***variance risk premium is the name we gave to the fact that option prices are usually expensive***.

VRP exists due to several factors that influence the pricing and demand for options. These factors create a consistent discrepancy between implied and realized volatility, resulting in a premium that traders can exploit.

The reason we care so much about the VRP is that it’s the reason we even consider selling options as a way to make money. If there was no phenomenon to build off of, then there would be no reason that selling options should continue to be profitable in the future.

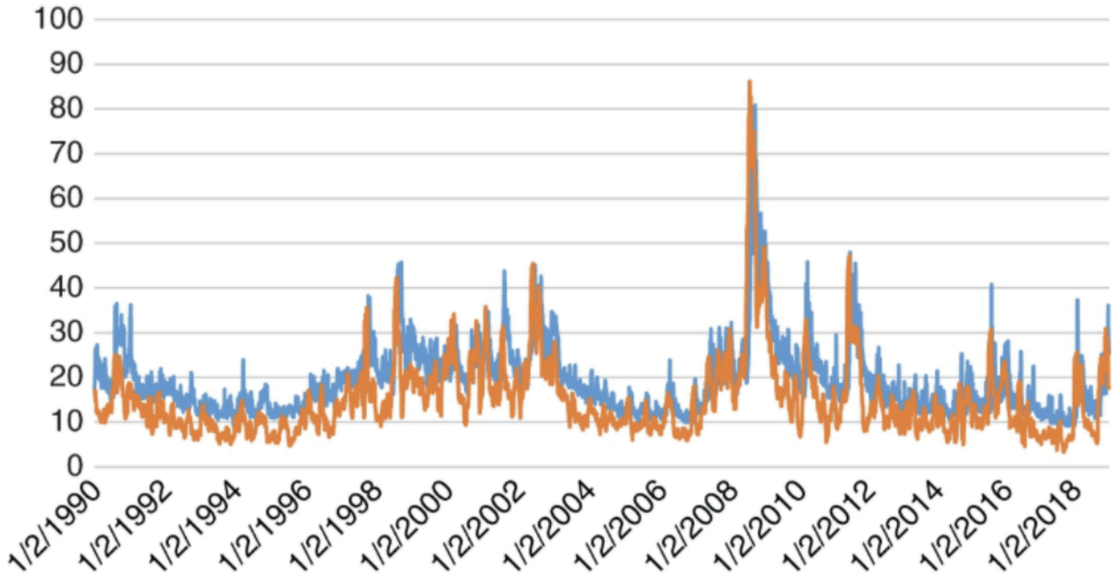
Thankfully, it does exist and will continue to exist. Let’s prove why.

Let's start with some data.

Before anything, let's look at some hard numbers that show what we are talking about actually exists.

The S&P 500 is a good example to look at

The graph below shows the difference between implied volatility and realized volatility for the S&P 500. Since the S&P is often used to represent the market, this graph is used to demonstrate the existence of the variance risk premium.



The blue line is the implied volatility and the orange line is the realized volatility. The graph shows us that on average the implied volatility was **4 points higher** than the realized volatility and that the premium is **positive 85% of the time.** To help you understand what this means, an example would be that if IV was 20% on average, then RV would be 16% on average, resulting in a 4% variance risk premium for sellers. The graph also shows that the variance risk premium has existed for decades, and for reasons we discuss below, it's likely to continue.

What about the VRP for other indexes?

The same analysis was run for Dow Jones, NASDAQ 100, and Russell 2000 and the table below presents the key statistical findings.

Index	Dow Jones (from 1998)	NASDAQ 100 (from 2001)	Russell 2000 (from 2004)
Mean	3.50	3.41	3.24
Standard deviation	6.18	6.99	6.58
Skewness	−2.03	−2.02	−2.72
Maximum	28.90	26.48	27.28
Minimum	−49.42	−53.79	−46.08
Median	4.22	3.91	4.05
90th percentile	9.46	10.37	9.10
10th percentile	−2.60	−3.11	−2.81

In the above table, we can see that the mean VRP for each index is around 3 to 3.5. They share a similar statistical profile across the board. For all major indexes, the idea of a variance risk premium can be observed, and it is the entire reason we sell options.

This phenomenon, which has persisted for decades, is the backbone of profitable option trading. Billions of dollars have been raised and traded based on this core idea. Your new strategies will be too.

So why does VRP exist?

Data alone isn't enough because it looks backwards. To build confidence that it will continue into the future we want to be able to explain it with obvious and persistent logic that is tied to clear reasons it should make money.

There are four key reasons for the existence of VRP:

(1) Traders are willing to pay for insurance

The most compelling reason for a variance premium is that people are willing to pay for insurance. Diversification of a stock portfolio often fails during market crashes, and puts are the only reliable form of protection. Therefore, most of the VRP is driven by put buying.

(2) Traders are willing to pay to gamble

Calls can be mispriced due to fear of missing out (FOMO) events. Traders looking for exponential upside returns are not price sensitive to the call option they buy today ("*If it's going to jump 1,000%, who cares if the call option was $5 or $10!*" - Some retail trader)

(3) Options protect against sudden price jumps

Underlying prices can experience sudden jumps, and options provide protection against these jumps in a way that a dynamic hedging scheme cannot, making options more attractive to buyers, whether they are hedgers or speculators.

(4) Most traders can only buy, not sell options

Many retail traders face restrictions from their brokers, such as being forbidden from selling naked options. This means an entire class of speculators can only be long volatility, which drives the variance premium higher.

Typically we only need one solid reason to understand a premium. In this case we have four. Any one of these factors is a good enough reason for it to exist, and it's unlikely that they will all disappear at once regardless of market changes. This gives us confidence that the risk premium will continue to be present on average in the future.

Calculating the Variance Risk Premium

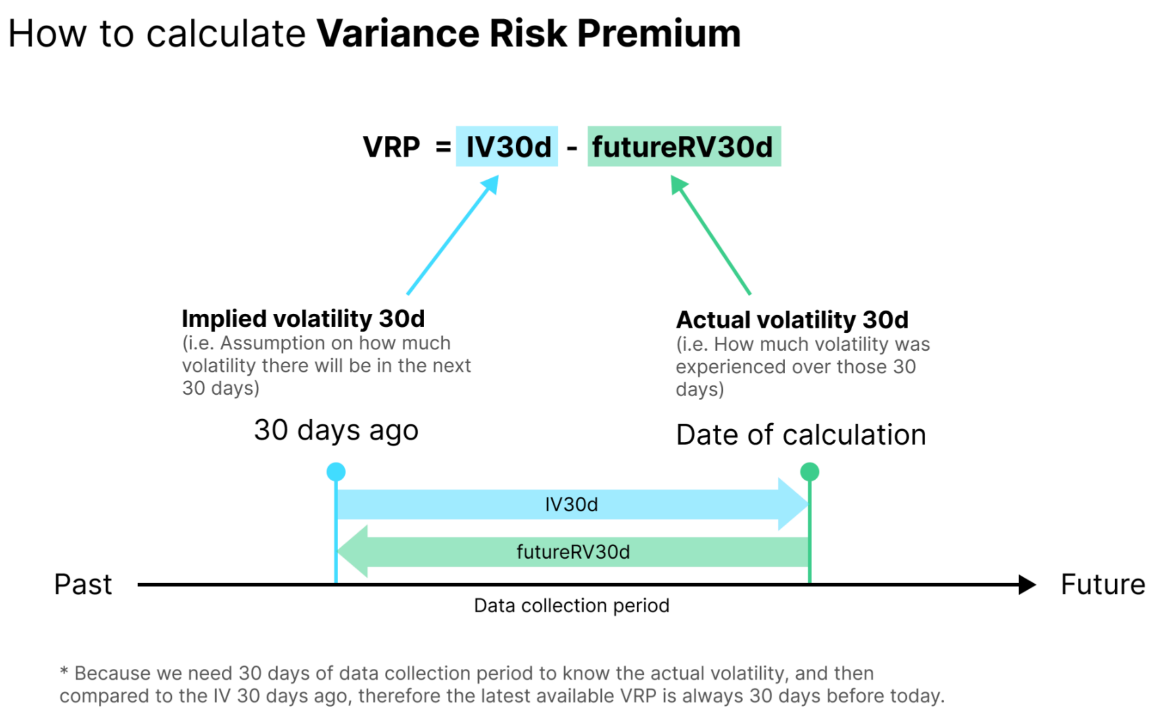
The strategies that we are going to be running are based on capturing the VRP. The reasons we trade, the tools we use, everything is based on this one concept and a lot of work goes into running these calculations.

Even though the platform handles the analytics for you, it's important to know how it works so you can trust the outputs you are using.

Here's how we calculate the variance risk premium:

Step 1: Compare Implied Volatility with Realized Volatility

To create a data point for our VRP model, we compare the market implied volatility for a period against the realized volatility experienced in that same period. We use the IV30d and the subsequent realized volatility over the next 30 days to determine the variance risk premium embedded on a given day.



Step 2: Create Data Points Over the Last 4 Years

One day's data isn't enough to make a trade. A single day shows if there was a profitable trade, but it doesn't tell us if there is an embedded premium for us to monetize. To determine this, we run this calculation for every day over a large period (4 years) and build a chart of the variance risk premium.

Remember the analysis that we did for the S&P 500 at the start of this section?

This tool basically allows us to do that analysis for any ticker in seconds. We can immediately see if the variance risk premium exists on an ETF or stock.

Examining the Variance Risk Premium

Once we have created this analysis, we extract 5 key metrics which help paint a clear picture of the premium landscape for an ETF.

The main metric:

Average VRP: The main metric that we are interested in for the VRP analysis is the Avg. VRP. This is the average spread between implied volatility (30 day) and the subsequent realized volatility. This is calculated using the last 4 years of data. The reason this metric is so valuable is because we want to be trading tickers that have an established variance risk premium. Seeing a positive average VRP confirms that our hypothesis that a VRP exists for a ticker is true (implied volatility outpaces realized volatility on average) and that the VRP is likely to persist into the future.

Supporting metrics:

1. **Latest VRP**: This is the variance risk premium for the most recent 30 day window. It shows you what the last recorded data point for the variance risk premium is.
2. **VRP Frequency**: This shows you the number of days over the last 4 years where the variance risk premium was present in the options for the ticker you are analyzing. This is valuable because we want to be selling premiums that follow a typical “short volatility” profile, meaning that we should see a high number of days where there was a positive VRP.
3. **Median Option Volume**: This is the median number of option contracts traded daily over the last 4 years. It’s valuable because it indicates to us the level of liquidity in the option chain over the last 4 years. We want to make sure that there is a reasonable level of liquidity because it means that the data we are able to pull from the option chain is going to be more accurate.

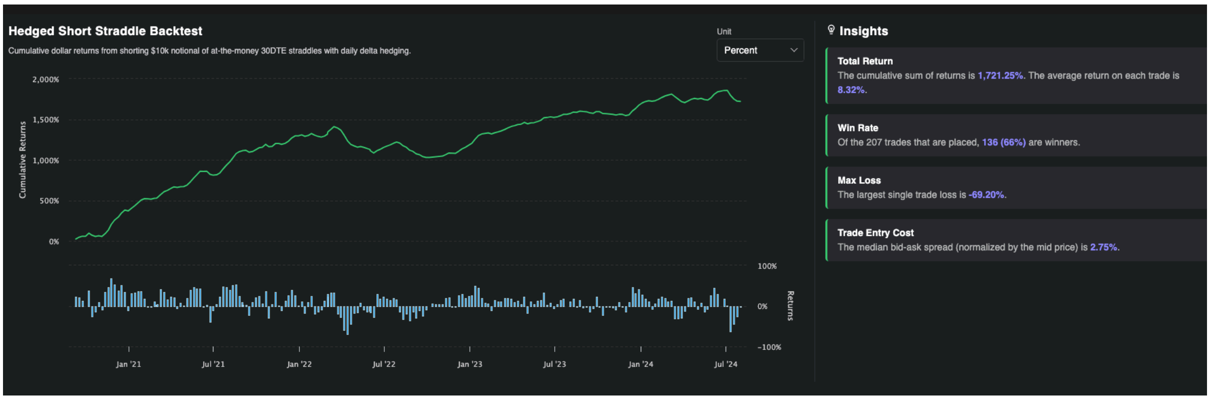
At a glance you can measure the size of a risk premium for an asset.

Pretty cool, huh?

This VRP analysis is available to you in the terminal, between this and the option selling backtests we generate, you get to see a full picture of the risk premium available for a ticker, and whether it is something we can monetize.

Confirming if the Variance Risk Premium Can Be Monetized: Backtests

Knowing that the variance risk premium exists and being able to capture it are two different things. Remember, as retail traders we need to be able to place trades that actually make money. Knowing the variance risk premium exists is the most important part, but once we know this we will want to take things one step further and run a backtest to see if we are able to actually make money and harvest this variance risk premium after accounting for transaction costs, delta hedging, and slippage.



Built into the Predicting Alpha terminal is a backtester for both ETFs and earnings events (both of the strategies you will be learning).

The image above is for the backtest on ETFs, in particular on $DIA in the picture.

What goes into this backtest?

The way the backtest works is by finding every time a 30 DTE contract was available to trade over the last 4 years (keeping it in line with our VRP analysis), selling a straddle, delta hedging it daily, and seeing what the performance looks like. The built-in assumption for costs is that you must cross the bid/ask spread completely in order to enter the trade.

Examining the backtest

The main metric:

The main metric that we are interested in for the backtest is the average straddle return. You can think of this as the *practical average variance risk premium*. It's the average return per trade over the 4 year period that was generated from selling options.

Supporting Metrics:

1. **Total return**: the sum return of all the trades taken. It is important to note that the total return is the sum of all the straddle PnLs and not the return on your portfolio.
2. **Win Rate**: This tells you how many of the trades placed were profitable. Some traders will find this useful for being able to understand their risk profile on the trade. When you are in a trade you want to make sure that what you are seeing in real time is a reflection of your expectations. Having an idea for how often you should be winning helps you measure how things are going.
3. **Max Loss**: This tells you the biggest drawdown taken on any trade in the backtest. This metric is really valuable when determining how big to size your position - you want to make sure you can absorb potential drawdowns and stay in the game.
   A**Trade Entry Cost**: This tells you the average bid ask spread, which is useful for knowing how much you are paying in slippage if you choose to trade this ticker. Note that the bid ask spread is normalized by the mid price.

Change between percentages and dollars

The last detail you need to know about the backtesting available in the platform is that you are able to convert it between percentages and dollar returns. If you wanted to view the backtests in terms of dollars, it assumes that you allocated $10,000 to each trade. You need to know this to understand the dollar amounts you are seeing.

Seeing the full picture: VRP + backtest

By looking at the data from the VRP calculation and the backtest, we are able to paint a full picture. We can see if the variance risk premium exists for a given ticker and if we can actually monetize it after considering costs and delta hedging.

I will consider this chapter a success if you:

1) Have a really clear understanding of what the variance risk premium is and why it exists.
2) Understand how it is calculated.
3) Understand how the VRP chart and the backtest work together to paint a really clear picture about if there is a variance risk premium that we can monetize on a ticker.

Do you feel confident in these three things? If so, great!

If you have any questions, use the chat box in the bottom right corner to send me a message. I'll personally help you out!

You have reached the end of the theory section. In the next chapter, we are going to learn the ETF Premium strategy!